JAGADISH BUTUKURI

Senior Data Engineer

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PROFESSIONAL SUMMARY

- Senior Data Engineer with 5+ years designing, building, and operating enterprise data and AI platforms on AWS, Azure, Databricks, and Snowflake for agribusiness, healthcare, and financial services.
- Built batch and real-time pipelines, lakehouse architectures, data models, integrations, and governed data products with Apache Spark, PySpark, Kafka, dbt, Delta Lake, and orchestration frameworks.
- Built production GenAI and RAG solutions using LangChain, OpenAI APIs, Pinecone vector databases, semantic/vector search, embeddings, MLflow, SageMaker, Feast, and Model Context Protocol (MCP), including tool-enabled LLM integrations with governed enterprise data.
- Strong Experience in data quality, observability, governance, security, infrastructure as code, CI/CD, platform reliability, analytics enablement, and cross-functional delivery in large-scale production environments.

PROFESSIONAL EXPERIENCE

Senior Data Engineer | Cargill | Minneapolis, MN **Sep 2024 – Present**

- Designed and built a cloud-native ELT platform on AWS and Databricks ingesting terabytes of supply-chain, procurement, commodity, and operational data daily; used Delta Lake and schema contracts to improve platform reliability and reduce stale-data incidents.
- Developed batch and real-time pipelines with PySpark, Spark Structured Streaming, Kafka Connect, and Confluent Schema Registry, replacing scheduled batch processing with near-real-time delivery for downstream consumers.
- Implemented dbt Core and dbt Mesh across commodity, procurement, and logistics domains, including semantic and metrics layers that improved governed self-service access for business teams.
- Built reusable data models, data contracts, and automated quality checks with Great Expectations and Monte Carlo, supporting freshness SLAs and trusted gold-layer datasets for analytics and AI use cases.
- Engineered entity-resolution logic in Python and SQL to deduplicate millions of supplier records across multiple source systems and improve downstream matching and join quality.
- Built production RAG pipelines over enterprise document collections using LangChain, OpenAI APIs, Pinecone vector databases, semantic/vector search, NER, keyword tagging, and embedding workflows; tracked model and fine-tuning experiments in MLflow.
- Integrated SageMaker, Feast, and MLflow into the AI data platform for model training, inference, feature serving, experiment tracking, and embedding/version lifecycle management across structured and unstructured commodity data.
- Migrated Airflow-based orchestration toward Dagster asset-based workflows and used AWS Step Functions and AWS Batch for complex commodity, logistics, and supply-chain processing.
- Integrated SaaS and database sources using Fivetran, Airbyte, AWS DMS, MySQL, and Amazon RDS, landing governed data in the S3 lakehouse for downstream Spark processing.
- Optimized PySpark workloads in Databricks through adaptive query execution, partitioning, caching, and cluster right-sizing, reducing runtime and generating more than $100K in annual platform savings.
- Provisioned AWS services via Terraform, applied IAM, Lake Formation, CI/CD, Datadog monitoring, and FinOps practices; documented architectures and runbooks and led stakeholder reviews.
- Developed MCP servers and standardized LLM tool interfaces that let AI applications securely discover and query governed Delta Lake data, supporting reusable RAG, tool-enabled, and agent-ready workflows while preserving enterprise access controls.

Data Engineer | Molina Healthcare | Long Beach, CA **Jan 2021 – Jul 2022**

- Built healthcare data integration pipelines with Azure Data Factory, Synapse Analytics, and Databricks across member, provider, claims, pharmacy, and vendor datasets, improving consistency through MDM standards.
- Implemented HL7 FHIR-compliant ingestion and transformation for real-time patient events using Azure Event Hubs and PySpark on Databricks, supporting clinical decision dashboards and operational reporting.
- Configured CDC-based incremental loading with Debezium patterns, cutting full-refresh processing from ~6 hours to <1 hour and improving reporting freshness.
- Designed dimensional models and Data Vault 2.0 structures in ERwin for managed-care reporting, applying MDM to improve master-data consistency across clinical domains.
- Implemented HIPAA-aligned controls using Azure Key Vault, Azure Purview, RBAC, and PHI de-identification to meet healthcare data compliance.
- Established Azure Monitor and Soda validation for pipeline observability, SLA tracking, and schema-quality checks across ADF and Synapse workflows.
- Maintained pipeline runbooks and architecture documentation, handled incidents and changes through ServiceNow and Azure DevOps, and developed Power BI datasets for clinical operations and compliance teams.

Data Engineer | JPMorgan Chase | New York, NY **Dec 2018 – Dec 2020**

- Built Spark and Databricks pipelines on AWS processing hundreds of millions of daily retail-banking transaction records for multiple downstream teams and production reporting workloads.
- Developed ETL workflows in Informatica PowerCenter and Python to move Oracle data into S3 and Redshift, and optimized SQL transformations to reduce query runtime and recurring batch failures.
- Migrated a legacy SQL Server warehouse to Snowflake for analytics and regulatory reporting and used Presto-based federated queries to improve access across distributed data sources.
- Built real-time event streams with Kafka Connect and Confluent Schema Registry, enforcing producer contracts and routing high-volume events for downstream banking and fraud analytics.

- Stored fraud-signal data in Cassandra and Redis for low-latency risk-scoring workflows and supported secure access patterns for sensitive customer data.
- Provisioned AWS data infrastructure with CloudFormation and CDK and standardized configuration across development, staging, and production environments; supported IAM, encryption, and containerized Kubernetes deployments.
- Partnered with risk, compliance, and regulatory reporting teams on data SLAs, incident resolution, code reviews, and technical documentation for enterprise consumers.

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, PL/SQL, Scala, Shell scripting, Java
Data Engineering & Lakehouse: Apache Spark, Spark Structured Streaming, Databricks, Delta Lake, Apache Iceberg, Hadoop, Hive, Snowpark, Medallion architecture
Cloud Platforms: AWS (S3, EMR, Glue, Lambda, Redshift, RDS, DMS, MSK, SageMaker, Step Functions, Lake Formation, IAM); Azure (ADF, ADLS Gen2, Synapse, Azure SQL, Event Hubs, Stream Analytics, Purview, Monitor, Key Vault, Azure Databricks)
Streaming & Integration: Apache Kafka, Kafka Connect, Confluent Schema Registry, Apache Flink, Apache Beam, Debezium, Fivetran, Airbyte, Informatica PowerCenter, AWS DMS
Modeling, Quality & Governance: dbt Core/Cloud/Mesh, semantic and metrics layers, dimensional modeling, Data Vault 2.0, star schema, MDM, ERwin, Great Expectations, Soda, Monte Carlo, data contracts, lineage and freshness monitoring
Databases & Warehouses: Snowflake, Redshift, Synapse, PostgreSQL, MySQL, Oracle, SQL Server, Cassandra, Redis, DynamoDB, MongoDB, Elasticsearch
Orchestration, DevOps & IaC: Airflow, Dagster, Databricks Workflows, AWS Step Functions, Terraform, CDK, CloudFormation, GitHub Actions, GitLab CI, Jenkins, Docker, Kubernetes
GenAI, RAG & AI Platforms: RAG, LangChain, OpenAI APIs, Pinecone/vector databases, semantic and vector search, embeddings and embedding versioning, Model Context Protocol (MCP), LLM tool integration, agent-ready workflows, MLflow, SageMaker, Feast, LLM fine-tuning, Spark MLlib
BI, Security & Delivery: Power BI, Tableau, Looker, Grafana, IAM, RBAC, HIPAA, HL7/FHIR, Unity Catalog, Lake Formation, Azure Purview, Jira, Confluence, SharePoint, ServiceNow, Agile/Scrum

CERTIFICATIONS

AWS Certified AI Practitioner | Amazon Web Services

EDUCATION

Master of Science in Information Systems | Saint Louis University May 2024